

Kalyani Kansal

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Education

- **PhD in Mathematics**, Johns Hopkins University, USA (2016–2023). Advisor: David Savitt. Thesis: *Codimension one intersections between components of the Emerton-Gee stack for GL_2* .
- **MBBS (Bachelor of Medicine and Bachelor of Surgery)**, Government Medical College, Chandigarh, India (2007–2013).

Appointments

- **Chapman Fellow**, Imperial College London, UK (2024–).
- **Member, School of Mathematics**, Institute for Advanced Study, Princeton, USA (2023–2024).
- **Post-doctoral Fellow in Psychiatry**, Johns Hopkins University School of Medicine, USA (2013–2016).

Research Interests

Number theory, Arithmetic geometry.

(Pre-)Publications

- K. Kansal, B. Levin, D. Savitt. “Towards mod p local-global compatibility for partial weight one Hilbert modular forms.” *In preparation*.
- K. Kansal, B. Savoie. “Non-generic components of the Emerton-Gee stack for GL_2 ”. *Submitted*.
- R. Bellovin, N. Borade, A. Hilado, K. Kansal, H. Lee, B. Levin, D. Savitt, H. Wiersema. “Irregular loci in the Emerton-Gee stack for GL_2 ”. *Journal für die reine und angewandte Mathematik*, 2024.
- K. Kansal. “Codimension one intersections between components of the Emerton-Gee stack for GL_2 ”. To appear in *Trans. Amer. Math. Soc.*
- A. Guzman, K. Kansal, I. Kountouridis, B. Savoie, X. Wang. “Smoothness of components of the Emerton-Gee stack for GL_2 ”. *Trans. Amer. Math. Soc.*, 2025.

Invited Talks

- IHES Conference on “New Structures and Techniques in p-adic Geometry”, Bures-sur-Yvette, France (2025)
- BIRS Workshop on “Categorical and geometric structures in the Langlands program”, Oaxaca, Mexico (2025).
- Arbeitsgemeinschaft on “Relative Langlands Duality”, Oberwolfach, Germany (2025).
- Durham University Arithmetic Study Group, Durham, UK (2025).
- University of Glasgow Number Theory Seminar, Glasgow, UK (2025).
- University of Oxford Number Theory Seminar, Oxford, UK (2024).

- London Number Theory Seminar, London, UK (2024).
- Program for Women+ and Mathematics, Institute for Advanced Study, Princeton, USA (2024).
- New York Number Theory Seminar, New York, USA (2024).
- University of Chicago Number Theory Seminar, Chicago, USA (2023).
- Quebec–Vermont Number Theory Seminar, Montréal, Canada (2023).
- Cambridge Number Theory Seminar, Cambridge, UK (2023, online).
- Universität Heidelberg Number Theory Seminar, Heidelberg, Germany (2023).
- Hausdorff Trimester on “The Arithmetic of the Langlands Program”, Bonn, Germany (2023).
- Philadelphia Area Number Theory Seminar, Philadelphia, USA (2023).
- Essen Number Theory Seminar, Essen, Germany (2023, online).
- IAS/Princeton Arithmetic Geometry Seminar, Princeton, USA (2023).
- AMS Fall Eastern Sectional Meeting, USA (2023, online).
- Joint Mathematical Meetings, AMS Special Session on Women in Automorphic Forms, Boston, USA (2023).
- Johns Hopkins Junior Number Theory Days, Baltimore, USA (2022).
- Community Building in the Langlands Program, Bonn, Germany (2022).
- UC San Diego Number Theory Seminar, San Diego, USA (2022).
- Rice University AGNT Seminar, Houston, USA (2022).
- Ohio State Number Theory Seminar, Columbus, USA (2022, online).

Other Conferences and Special Programs Attended

- Simons Collaboration Annual Meeting on “Perfection in Algebra, Geometry and Topology”, New York, USA (2025, 2024).
- Clay Research Conference and Workshop on “Geometry and Langlands”, Oxford, UK (2024).
- Conference on “Representations of p -adic Groups and Langlands Correspondence” (in honor of Colin Bushnell), London, UK (2024).
- Oberwolfach Meeting on Arithmetic Geometry, Oberwolfach, Germany (2024).
- “Visions in Arithmetic and Beyond” (Peter Sarnak’s birthday conference), Princeton, USA (2024).
- IAS Special Year on p -adic Arithmetic Geometry, Princeton, USA (2023–2024).
- Spring School on Non-Archimedean Geometry and Eigenvarieties, Heidelberg, Germany (2024).
- Hausdorff Trimester on “The Arithmetic of the Langlands Program”, Bonn, Germany (2023).
- Collaborative Research Workshop, Eugene, USA (2022).
- Arizona Winter School: Automorphic Forms Beyond GL_n , Tucson, USA (2022).
- Community Building in the Langlands Program, Bonn, Germany (2022).
- MSRI Summer School: Toric Varieties, Taipei, Taiwan (2019).
- Workshop on Serre Conjectures and p -adic Langlands, Padova, Italy (2019).
- MSRI Summer School: Derived Categories, Berkeley, USA (2018).

Teaching

- **Instructor (Imperial College London):** Group Theory (Final-year undergraduates, ongoing).
- **Project Supervisor (Imperial College London):** Year 1 undergraduate cohort projects (~50 students, 2025); Final-year undergraduate independent study projects (2 students, 2025).
- **Instructor (Johns Hopkins):** Linear Algebra (2017, 2018); Introduction to Calculus (2021).
- **Teaching Assistant (Johns Hopkins):** Calculus I–III, Linear Algebra, Complex Analysis, Abstract Algebra (2016–2022).

Service

- Co-organiser, study group on “Positivity criteria for automorphic vector bundles on unitary Shimura varieties” (ongoing).
- Co-organiser, London Number Theory Seminar (2024); Hausdorff Trimester Seminar (2023).
- Referee: *Proc. Amer. Math. Soc.*, *Math. Z.*, and others.

Selected non-Math Publications

- K. Kansal et al. “Structural cerebellar correlates of cognitive and motor dysfunctions in cerebellar degeneration.” *Brain*, 2017.
- K. Kansal et al. “Survival in frontotemporal dementia phenotypes: a meta-analysis.” *Dementia and Geriatric Cognitive Disorders*, 2016.
- K. Kansal, D. J. Irwin. “The Use of Cerebrospinal Fluid and Neuropathologic Studies in Neuropsychiatry Practice and Research.” *Psychiatric Clinics*, 2015.